

Hot Pair Coulomb Density Matrix on a Sphere

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We tabulate the hot diagonal pair Coulomb density matrix on a sphere.

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I. INTRODUCTION

We show a method to lower the temperature of the hot Pair Coulomb Density Matrix (PCDM) on a Sphere so to extend the work of Ref. [1–3].

We will denote with a Greek index a coordinate basis component and with a Roman index a particle label. We will adopt Einstein summation convention to tacitly assume a sum over repeated Greek indexes. We distinguish between upper contravariant Greek indexes and lower covariant Greek indexes. Passing between the two is accomplished with a contraction with the metric tensor, as usual.

The metric tensor of the Sphere of radius a is

$$||g_{\alpha\beta}|| = a^2 \begin{pmatrix} 1 & 0 \\ 0 & \cos^2 \theta \end{pmatrix}. \quad (1.1)$$

with $g(\mathbf{r}) = \det||g_{\alpha\beta}|| = a^4 \cos^2 \theta$.

It is important to realize that on a curved surface the two body problem is generally not separable into a center of mass and a relative one body problem. Therefore, in full generality, we will study the position representation $\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ of the density matrix $\rho = \exp(-Ht)$, where H is the Hamiltonian operator for the two bodies and t is the imaginary time that in statistical physics is identified with the inverse temperature $\beta = 1/k_B T$ with k_B Boltzmann constant. These are initially at $\mathbf{r}_1$ and $\mathbf{r}_2$ and at an imaginary time t later they are at $\mathbf{r}'_1$ and $\mathbf{r}'_2$.

Therefore, the PCDM on the Sphere will satisfy the Bloch equation

$$\frac{\partial \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}{\partial t} = \left(\frac{\lambda_1}{\sqrt{g_1}} \nabla_{1\alpha} \sqrt{g_1} \nabla_1^\alpha + \frac{\lambda_2}{\sqrt{g_2}} \nabla_{2\alpha} \sqrt{g_2} \nabla_2^\alpha - \phi(\mathbf{r}_1, \mathbf{r}_2) \right) \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t), \quad (1.2)$$

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; 0) = \delta(\mathbf{r}_1 - \mathbf{r}'_1) \delta(\mathbf{r}_2 - \mathbf{r}'_2) / \sqrt{g_1 g_2}, \quad (1.3)$$

where, for particle $i = 1, 2$, $\lambda_i = \hbar^2/2m_i$ with m_i the mass of particle i , $g_i = a^4 \sin^2 \theta_i$, $\nabla_i = (\nabla_{i1}, \nabla_{i2}) = (\partial/\partial\theta_i, \partial/\partial\varphi_i)$ is the gradient respect to the coordinates $\mathbf{r}_i = (r_i^1, r_i^2) = (\theta_i, \varphi_i)$, δ is a two dimensional Dirac delta function, ρ is the pair density matrix, and ϕ is the Coulomb pair potential.

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The Coulomb pair potential for particles moving **on** the surface can be chosen as

$$\phi_{\text{on}}(\mathbf{r}_1, \mathbf{r}_2) = Z_1 Z_2 e^2 / d(\mathbf{r}_1, \mathbf{r}_2), \quad (1.4)$$

where $Z_i e$ is the charge of particle i and for the distance between the two points $\mathbf{r}_1$ and $\mathbf{r}_2$, $d(\mathbf{r}_1, \mathbf{r}_2)$, on the Sphere we can either choose the geodesic distance

$$d_g(\mathbf{r}_1, \mathbf{r}_2) = a \arccos[\sin \theta_1 \sin \theta_2 + \cos \theta_1 \cos \theta_2 \cos(\varphi_1 - \varphi_2)], \quad (1.5)$$

or, viewing the Sphere as embedded in the three dimensional space, the Euclidean distance

$$d_e(\mathbf{r}_1, \mathbf{r}_2) = 2a \sin(d_g/2a). \quad (1.6)$$

For particles living **in** the surface [4] we may choose

$$\phi_{\text{in}}(\mathbf{r}_1, \mathbf{r}_2) = -Z_1 Z_2 e^2 \ln[d_e(\mathbf{r}_1, \mathbf{r}_2)/\ell], \quad (1.7)$$

with ℓ a constant with the dimensions of a distance. We will call ϕ_{on}^g , ϕ_{on}^e , ϕ_{in} the three cases just illustrated.

We will measure energy in units of the Hartree e^2/a_B , with e the charge of the electron and $a_B = \hbar^2/m_e e^2$ Bohr radius, with m_e the mass of the electron. And length in units of the Bohr radius a_B . We will call $\mu_i = m_i/m_e$ the reduced mass of particle i .

The free particles, $\phi \rightarrow 0$, pair density matrix can be determined from the work of Bastianelli and Corradini [5]. We will call their solution for the single particle i , $\rho_{i0}(\mathbf{r}_i, \mathbf{r}'_i; t)$. Therefore the pair free density matrix will be

$$\rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_{10}(\mathbf{r}_1, \mathbf{r}'_1; t) \rho_{20}(\mathbf{r}_2, \mathbf{r}'_2; t). \quad (1.8)$$

Then our PCDM can be written as

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) e^{-P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}, \quad (1.9)$$

where P is the *inter-action*.

Aim of this work is to tabulate $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ for the three Coulomb cases illustrated above, P_{on}^g , P_{on}^e , P_{in} . In the hot case, for small $t \ll 1$, for each of these cases, the *primitive approximation* [6]

$$P_{\text{primitive}} \approx t\phi, \quad (1.10)$$

holds.

Therefore, introducing a timestep τ , we can find the PDCM at all subsequent timesteps through the following convolution path integral

$$\begin{aligned} \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; n\tau) &= \int \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_{1,1}, \mathbf{r}_{2,1}; \tau) \rho(\mathbf{r}_{1,1}, \mathbf{r}_{2,1} | \mathbf{r}_{1,2}, \mathbf{r}_{2,2}; \tau) \dots \rho(\mathbf{r}_{1,n-1}, \mathbf{r}_{2,n-1} | \mathbf{r}'_1, \mathbf{r}'_2; \tau) \times \\ &\quad \prod_{k=1}^{n-1} \sqrt{g_{1,k} g_{2,k}} d\mathbf{r}_{1,k} d\mathbf{r}_{2,k}, \end{aligned} \quad (1.11)$$

where we denote with $\mathbf{r}_{i,k}$ the position of particle i at timeslice k , $g_{i,k}$ is the determinant g calculated at $\mathbf{r}_{i,k}$, $d\mathbf{r} = d\theta d\varphi$, and we choose τ small enough so that

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau) \approx \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2} \quad (1.12)$$

$$= \rho_{10}(\mathbf{r}_1, \mathbf{r}'_1; \tau) \rho_{20}(\mathbf{r}_2, \mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2}, \quad (1.13)$$

is a good approximation. According to the analysis of Ref. [5] we can further approximate

$$g^{1/4}(\mathbf{r}_i) \rho_{i0}(\mathbf{r}_i, \mathbf{r}'_i; \tau) g^{1/4}(\mathbf{r}'_i) \approx \exp \left\{ -\frac{1}{2\mu_i \tau} \delta_{\alpha\beta} (\mathbf{r}_i - \mathbf{r}'_i)^\alpha (\mathbf{r}_i - \mathbf{r}'_i)^\beta - \tau \frac{V_R(x_i) + V_R(x'_i)}{2} \right\}, \quad (1.14)$$

where $\delta_{\alpha\beta}$ is the Kronecker delta, $x_i^2 = \delta_{\alpha\beta} \mathbf{r}_i^\alpha \mathbf{r}_i^\beta$, and V_R is the effective potential due to the curvature of the Sphere, $R = 2M^2$ with $M = 1/a$, namely

$$V_R(x) = -\frac{M^2}{6} - \frac{5(Mx)^2 - 3 + [(Mx)^2 + 3] \cos(2Mx)}{48x^2 \sin^2(Mx)}, \quad (1.15)$$

which is equation (2.32) of Ref. [5]. We will perform the $4(n-1)$ integrations in Eq. (1.11) through a simple quadrature method.


```

real*8      th1,th2,ph1,ph2,th1p,th2p,ph1p,ph2p
real*8      dg,de,dgp,dep
real*8      rhoid,rho,pot,dth,dph,pcdm,fa,pidm,faid
integer*8    i,j,k,ni,na,nth,nph,id,nd
integer*8    ith1(mnd),iph1(mnd),ith2(mnd),iph2(mnd)
character    potk*3,tabf*11
parameter    (a=1.,tau=0.1,delta=0.3)
eps=1.d-3
pi=acos(-1.d0)

write(*,*) '.....'
write(*,*) 'Sphere radius      a =',a
write(*,*) 'time step          tau =',tau
write(*,*) 'g distance step delta =',delta
write(*,*) '.....'
write(*,*) 'm1 and m2 in electron mass units'
read (*,*) mu1,mu2
write(*,*) 'Z1 and Z2'
read (*,*) z1,z2
write(*,*) '# timestep iterations, n >= 1'
read (*,*) ni
ti =(ni+1)*tau
write(*,*) 'imaginary time t = (n+1)*tau =',ti
write(*,*) '# angular distance discretization points , m <=',mnd
read (*,*) na
da =pi/na
write(*,*) 'angle discretization step da = pi/m =',da
write(*,*) 'give the kind of Coulomb potential (in,one,ong)'
read (*,*) potk
tabf = 'tab-'//trim(potk)//'.dat'
open(unit=8,file=tabf,status='unknown')

dth = da
dph = 2*da
nd = int(pi/delta)

ith1=1
ith2=1
iph1=1
iph2=1
c loop over geodesic distance
do id=1,nd
    th10=id*delta-pi/2
    ph10=eps
    th20=-pi/2
    ph20=eps
    dg0=a*acos(sin(th10)*sin(th20)+cos(th10)*cos(th20)*
&      cos(ph10-ph20))
    de0=2*a*sin(dg0/2/a)
    pcdm = 0.
    pidm = 0.
c loop over timesteps
10 th1=ith1(1)*dth-pi/2
    ph1=iph1(1)*dph
    th2=ith2(1)*dth-pi/2
    ph2=iph2(1)*dph
    dg=a*acos(sin(th1)*sin(th2)+cos(th1)*cos(th2)*cos(ph1-ph2))
    de=2*a*sin(dg/2/a)
    if (dg.lt.eps)goto 20
    fa=rho(potk,tau,a,mu1,mu2,z1,z2,
&      th10,ph10,th20,ph20,th1,ph1,th2,ph2,dg,de,dg0,de0)
    faid=rhoid(tau,a,mu1,mu2,
&      th1,ph1,th2,ph2,th10,ph10,th20,ph20)
    do j=2,ni
        th1=ith1(j-1)*dth-pi/2
        ph1=iph1(j-1)*dph
        th2=ith2(j-1)*dth-pi/2
        ph2=iph2(j-1)*dph
        th1p=ith1(j)*dth-pi/2
        ph1p=iph1(j)*dph
        th2p=ith2(j)*dth-pi/2
        ph2p=iph2(j)*dph
        dg=a*acos(sin(th1)*sin(th2)+cos(th1)*cos(th2)*cos(ph1-ph2))
        de=2*a*sin(dg/2/a)
        dgp=a*acos(sin(th1p)*sin(th2p)+cos(th1p)*cos(th2p)*
&      cos(ph1p-ph2p))
        dep=2*a*sin(dgp/2/a)
        if (dg.lt.eps)goto 20
        if (dgp.lt.eps)goto 20
        fa=fa*dth**2.*dph**2.*rho(potk,tau,a,mu1,mu2,z1,z2,
&      th1,ph1,th2,ph2,th1p,ph1p,th2p,ph2p,dg,de,dgp,dep)
        faid=faid*dth**2.*dph**2.*rhoid(tau,a,mu1,mu2,
&      th1,ph1,th2,ph2,th1p,ph1p,th2p,ph2p)
    enddo
    th1=ith1(ni)*dth-pi/2
    ph1=iph1(ni)*dph
    th2=ith2(ni)*dth-pi/2
    ph2=iph2(ni)*dph
    dg=a*acos(sin(th1)*sin(th2)+cos(th1)*cos(th2)*
&      cos(ph1-ph2))
    de=2*a*sin(dg/2/a)
    if (dg.lt.eps)goto 20
    fa=fa*dth**2.*dph**2.*rho(potk,tau,a,mu1,mu2,z1,z2,
&      th1,ph1,th2,ph2,th10,ph10,th20,ph20,dg,de,dg0,de0)
    faid=faid*dth**2.*dph**2.*rhoid(tau,a,mu1,mu2,
&      th1,ph1,th2,ph2,th10,ph10,th20,ph20)
    pcdm = pcdm+fa
    pidm = pidm+faid
20 do k=1,ni
    ith1(k)=ith1(k)+1
    if (ith1(k).le.na) goto 10
    ith2(k)=ith2(k)+1
    ith1(k)=1
    if (ith2(k).le.na) goto 10
    iph1(k)=iph1(k)+1
    ith1(k)=1
    ith2(k)=1
    if (iph1(k).le.na) goto 10
    iph2(k)=iph2(k)+1
    ith1(k)=1
    ith2(k)=1

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```

        iph1(k)=1
        if(iph2(k).le.na) goto 10
        do j=1,k
            ith1(j)=1
            ith2(j)=1
            iph1(j)=1
            iph2(j)=1
        enddo
    enddo
c    tabulate P=-log(rho/rho0)
        pcdm = -log(pcdm/pidm)
        write(8,100) a*dg0,pcdm,ti*pot(potk,a,z1,z2,dg0,de0)
        write(*,*) a*dg0,pcdm,ti*pot(potk,a,z1,z2,dg0,de0)
    enddo

100 format(3(d10.3,3x))
close(unit=8)

stop
end

function fact(n)
implicit none
factorial function
real*8 fact,p
integer i,n

p=1.
do i=1,n
    p=p*dble(i)
enddo
fact=p

return
end

function vr(a,mu,th,ph)
implicit none
c Eq. (2.32) Bastianelli and Corradini EPJC 77, 731 (2017)
real*8 vr,a,mu,th,ph
real*8 pi,m,x,mx,eps
eps=1.d-3
pi=acos(-1.d0)

m =1/a
x =sqrt(th**2.+ph**2.)
mx=m*x
vr=-m**2./6-(5*m*x**2.-3*(mx**2.+3)*cos(2*mx))/48/x**2./sin(mx)**2.
if(abs(vr).gt.1/eps) vr=0.

return
end

function pot(potk,a,z1,z2,dg,de)
implicit none
c the Coulomb potential
real*8 pot,a,z1,z2
real*8 pi,de,dg
character potk*3
pi=acos(-1.d0)

if(potk.eq.'in') then
    pot=-z1*z2*log(de)
elseif(potk.eq.'one') then
    pot=z1*z2/de
elseif(potk.eq.'ong') then
    pot=z1*z2/dg
else
    write(*,*) 'no pair-potential available !'
    stop
endif

return
end

function rho(potk,tau,a,mu1,mu2,z1,z2,
& th1,ph1,th2,ph2,thip,phip,th2p,ph2p,dg,de,dgp,dep)
implicit none
c primitive approximation for rho
real*8 rho,tau,a,mu1,mu2,z1,z2,dg,de,dgp,dep
real*8 th1,ph1,th2,ph2,thip,phip,th2p,ph2p
real*8 pot,vr,ddx1,ddx2
character potk*3

ddx1=(th1-thip)**2.+(ph1-phip)**2.
ddx2=(th2-th2p)**2.+(ph2-ph2p)**2.

rho = exp(-ddx1/2/mu1/tau-tau*
& (vr(a,mu1,th1,ph1)+vr(a,mu1,thip,phip))/2)
rho =rho*exp(-ddx2/2/mu2/tau-tau*
& (vr(a,mu2,th2,ph2)+vr(a,mu2,th2p,ph2p))/2)
rho =rho*exp(-tau*(pot(potk,a,z1,z2,dg,de)+
& pot(potk,a,z1,z2,dgp,dep))/2)

return
end

function rhoid(tau,a,mu1,mu2,
& th1,ph1,th2,ph2,thip,phip,th2p,ph2p)
implicit none
c primitive approximation for rho
real*8 rhoid,tau,a,mu1,mu2
real*8 th1,ph1,th2,ph2,thip,phip,th2p,ph2p
real*8 vr,ddx1,ddx2

ddx1=(th1-thip)**2.+(ph1-phip)**2.

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ddx2=(th2-th2p)**2.*(ph2-ph2p)**2.

rhoid =      exp(-ddx1/2/mu1/tau-tau*
&      (vr(a,mu1,th1,ph1)*vr(a,mu1,th1p,ph1p))/2)
rhoid =rhoid*exp(-ddx2/2/mu2/tau-tau*
&      (vr(a,mu2,th2,ph2)*vr(a,mu2,th2p,ph2p))/2)

return
end

```

Appendix A: $\rho_{i0}(\mathbf{0}, \mathbf{0}; t)$

Calling $\mu_i = m_i/m_e$ and $R = 2/a^2$ the Sphere scalar curvature, we find from equation (3.23) of Ref. [5]

$$\ln \left[2\pi t \sqrt{g(\mathbf{0})} \rho_{i,0}(\mathbf{0}, \mathbf{0}; t) \right] = \frac{tR_i}{12} + \frac{(tR_i)^2}{6!} \frac{1}{2} + \frac{(tR_i)^3}{9!} 16 + \frac{(tR_i)^4}{10!} \frac{59}{4} + \frac{(tR_i)^5}{11!} \frac{58}{3} + \frac{(tR_i)^6}{13!} \frac{550789}{1260} + \frac{(tR_i)^7}{14!} \frac{46838}{45} + \frac{(tR_i)^8}{17!} \frac{596004707}{720} + \mathcal{O}(t^9), \quad (\text{A1})$$

where $R_i = R/\mu_i$.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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